

Standard Operating Procedure (SOP)

Krauss Maffei 4000-24500 MX – Injection Molding Machine

1. Purpose

To provide a safe and standardized procedure for the setup, operation, and shutdown of the Krauss Maffei 4000-24500 MX injection molding machine. This SOP ensures operator safety, machine integrity, and high-quality output.

2. Scope

This SOP is mandatory for all trained personnel operating or working around the Krauss Maffei 4000-24500 MX machine at Integrity Tool & Mold Inc.

3. Responsibilities

- **Operators** are responsible for following this SOP and reporting any safety issues.
 - **Leaders** must verify operator certification and machine readiness.
 - **Maintenance Technicians** must perform regular inspections and preventive maintenance.
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4. Required PPE (Personal Protective Equipment)

All operators and personnel must wear:

- CSA-approved **safety glasses** or **face shield**
- **Steel-toe boots**
- **Heat-resistant gloves** (especially during mold changes or material purging)
- **Long-sleeved flame-resistant clothing**
- **Hearing protection**
- **Respiratory protection** (if fumes or fine particulates are present)

5. Pre-Operational Safety Checks

Before machine startup:

- Conduct a **360° walkaround inspection** for debris, leaks, or obstructions.
 - Check that **all guards, doors, and safety interlocks** are functional and NOT bypassed.
 - Inspect **hydraulic hoses, cooling lines, and electrical cables** for damage.
 - Ensure the **emergency stop buttons** function and are easily accessible.
 - Confirm the **mold is securely clamped** and properly aligned.
 - Verify the **cooling system** is active and flow is adequate.
 - Check that the **resin hopper** is loaded with the correct material.
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6. Machine Start-Up Procedure

1. Power on the main disconnect and HMI interface.
 2. Home all machine axes (robot) and initialize the system.
 3. Load the correct mold recipe/program from the approved setup sheet.
 4. Preheat the barrel and mold to designated temperatures.
 5. Ensure that the **clamp, injection, and ejection parameters** match the work order.
 6. Run a **dry cycle** to confirm functionality of:
 - Clamp movement
 - Injection unit
 - Ejector system
 - Core pulls (if applicable)
 7. Perform a trial shot and inspect for part quality and safety clearance.
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7. Operating Procedure

1. Confirm that the **safety gate is closed** and locked before starting the cycle.
2. Start automatic production cycle.
3. Remain at the operator panel and **monitor**:
 - Clamp force and mold protection
 - Injection pressure, speed, and cushion
 - Ejection timing and part removal
 - Cooling system temperature and flow

- Cycle times and alarm status
 - 4. If alarms occur, **stop the cycle** and notify a Leader immediately.
 - 5. Do not reach into the machine while it is cycling, even with interlocks.
 - 6. Use a robotic or mechanical arm for part removal whenever possible.
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8. Shutdown Procedure

1. Stop the production cycle and allow the final part to eject.
 2. Turn off barrel heaters and purge remaining material.
 3. Move the mold to an open position.
 4. Power down the machine using the proper shutdown sequence on the HMI.
 5. Lock out/tag out (LOTO) the machine if maintenance or cleaning is required.
 6. Clean up resin spills, wipe down the area, and document any issues.
 7. Log downtime or maintenance needs in the machine's logbook.
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9. Safety Protocols

- **Lockout/Tagout (LOTO):** Required for maintenance, mold changes, or tool cleaning.
 - **Emergency Stops:** Know all E-stop button locations and test regularly.
 - **High-Pressure Awareness:** Maintain safe distance from hydraulic and injection areas.
 - **No Bypass Policy:** Never override safety interlocks or doors.
 - **Burn Risk:** Avoid contact with barrel, hot runner, and nozzle during and after operation.
 - **Pinch Points:** Stay clear of the clamp area during mold open/close cycles.
 - **Material Handling:** Follow MSDS instructions and use safe lifting techniques.
 - **Fire Safety:** Keep flammable materials away from heating elements. Fire extinguisher must be accessible.
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10. Training Requirements

All operators must complete:

- Krauss Maffei 4000-24500 MX SOP training
- Lockout/Tagout (LOTO) certification
- WHMIS training for resins used